Credit Card Default Auto-ML Report

🕐 2026-05-16 05:19:33 UTC | 🏆 Best: **catboost** roc_auc = 0.7790 | 학습 24000행 | 테스트 6000행 | Feature 19개

1 모델 비교 (holdout 기준)

model	roc_auc	ks	lift	accuracy	precision	recall	f1	pr_auc	best_iter (avg)
lgbm	0.7748	0.4271	3.1424	0.7917	0.7939	0.0784	0.1427	0.5468	198
xgb	0.7773	0.4233	3.1575	0.8182	0.6720	0.3474	0.4580	0.5534	179
catboost BEST	**0.7790**	**0.4340**	**3.1198**	**0.8178**	**0.6590**	**0.3655**	**0.4702**	**0.5594**	**601**

2 CV (OOF) 비교

model	roc_auc	ks	lift	accuracy	precision	recall	f1	pr_auc
lgbm	0.7816	0.4327	3.1814	0.8201	0.6955	0.3321	0.4495	0.5548
xgb	0.7841	0.4349	3.1889	0.8215	0.6829	0.3601	0.4716	0.5580
catboost BEST	**0.7839**	**0.4353**	**3.1946**	**0.8226**	**0.6790**	**0.3758**	**0.4838**	**0.5601**

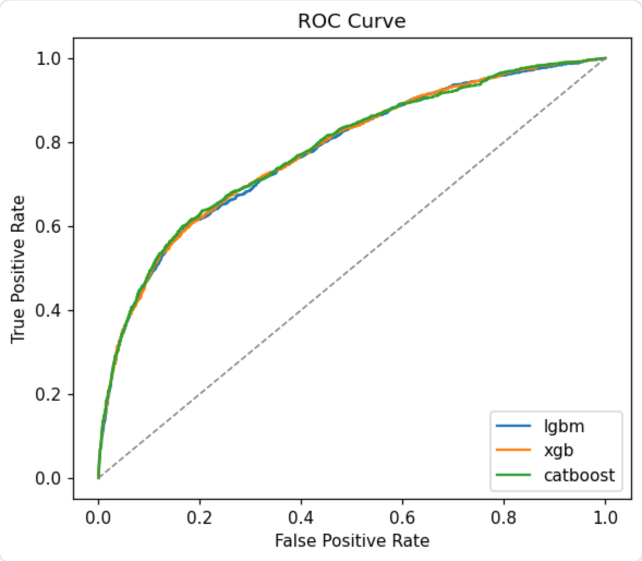
3 오버핏 점검 (Train vs Holdout)

model	roc_auc			ks			lift				
	Train	Test	Δ	Train	Test	Δ	Train	Test	Δ	Train	Te
lgbm	0.8268	0.7748	+0.0520	0.5026	0.4271	+0.0754	3.2360	3.1424	+0.0936	0.7910	0.791
xgb	0.8138	0.7773	+0.0365	0.4758	0.4233	+0.0525	3.3754	3.1575	+0.2179	0.8295	0.818
catboost BEST	**0.8077**	**0.7790**	+0.0287	**0.4624**	**0.4340**	+0.0284	**3.3490**	**3.1198**	+0.2292	**0.8287**	**0.817**

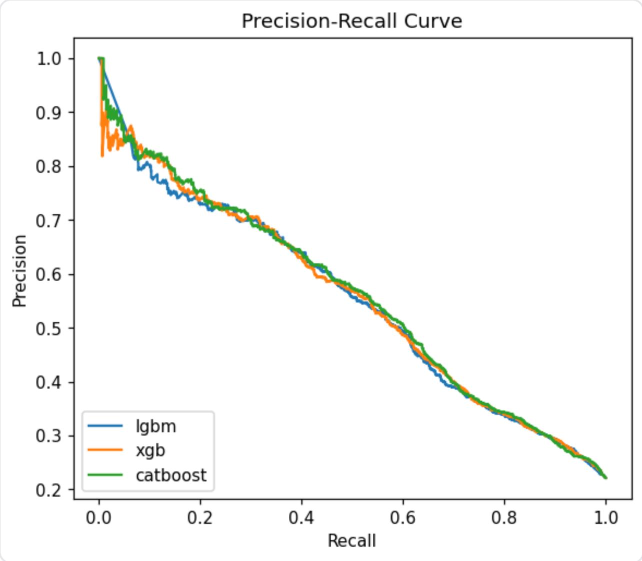
Δ = Train − Test. Train 은 final 모델을 학습 데이터에 fit 한 뒤 동일 학습 데이터로 in-sample 스코어링한 결과로, 의도적으로 over-optimistic 한 추정치다. Test 와의 양의 gap 이 클수록 메모리제이션(overfit) 신호이며, |Δ| ≥ 0.05 / |Δ| ≥ 0.10 인 셀은 색으로 강조한다. (정확한 임계치는 도메인/지표에 따라 다르며 본 색은 참고용.)

4 곡선 비교

ROC Curve



Precision-Recall Curve



5 Best 모델 상세 — catboost

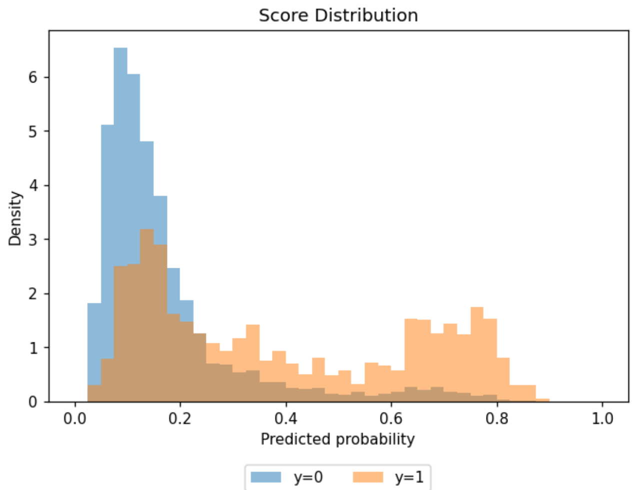
Feature Importance (top 30)

Rank	Feature	Importance
1	PAY_0	25.2403
2	LIMIT_BAL	9.8115
3	BILL_AMT1	8.1927
4	PAY_2	6.0274
5	PAY_3	4.4471
6	BILL_AMT2	4.3936
7	PAY_AMT2	4.3723
8	PAY_AMT3	4.2982
9	PAY_4	4.2650
10	PAY_AMT1	4.1239
11	AGE	3.5923
12	PAY_AMT4	3.0514
13	BILL_AMT5	3.0093
14	PAY_6	2.9744
15	PAY_5	2.9178

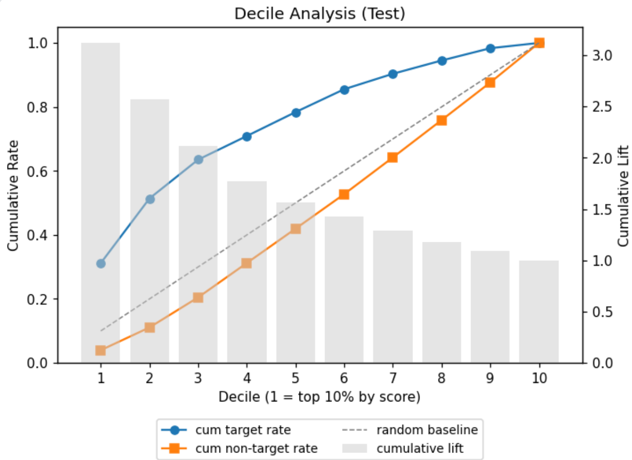
Rank	Feature	Importance
16	PAY_AMT6	2.6288
17	BILL_AMT6	2.5764
18	PAY_AMT5	2.5087
19	SEX	1.5690

10-분위 분석 (테스트셋, 점수 내림차순) 및 점수 분포

Score Distribution by Label



Decile Analysis Chart



Decile	Score Range	Count	Target	Non-Target	Cum Target	Cum NonTarget	Cum Count	Cum %
1	0.595 ~ 0.886	600	414	186	414	186	600	10.0%
2	0.320 ~ 0.594	600	268	332	682	518	1200	20.0%
3	0.214 ~ 0.320	600	161	439	843	957	1800	30.0%
4	0.170 ~ 0.213	600	98	502	941	1459	2400	40.0%
5	0.144 ~ 0.170	600	99	501	1040	1960	3000	50.0%
6	0.122 ~ 0.144	600	95	505	1135	2465	3600	60.0%
7	0.103 ~ 0.122	600	64	536	1199	3001	4200	70.0%
8	0.086 ~ 0.103	600	55	545	1254	3546	4800	80.0%
9	0.067 ~ 0.086	600	51	549	1305	4095	5400	90.0%
10	0.026 ~ 0.067	600	22	578	1327	4673	6000	100.0%

Confusion Matrix threshold = 0.5

	Pred 0	Pred 1
True 0	**4422**	251
True 1	842	**485**

6 학습 설정 요약

설정 키	값
`target_column`	`default`
`categorical_columns`	`SEX, EDUCATION, MARRIAGE`
`id_columns`	`client_id`
`preprocessing.numeric_null_strategy`	`median`
`preprocessing.categorical_null_strategy`	`most_frequent`
`preprocessing.outlier_method`	`percentile`
`preprocessing.outlier_action`	`clip`
`preprocessing.skew_method`	`signed_log1p`
`preprocessing.skew_threshold`	`1.0`
`preprocessing.scaling_method`	`standard`
`training.cv_folds`	`4`
`training.early_stopping_rounds`	`50`
`training.primary_metric`	`roc_auc`

설정 키	값
`training.random_state`	42
`tuning.enabled`	True
`tuning.n_trials`	15
`tuning.cv_folds`	3
`tuning.timeout`	None

7 변수 선택 (Stability Selection)

base: **lasso** | 부분표본 **80회** | 임계값 **0.60** | 채택 **19 / 23**

Feature	Selection Frequency	채택
LIMIT_BAL	100.0%	✓ 채택
AGE	100.0%	✓ 채택
PAY_0	100.0%	✓ 채택
PAY_2	100.0%	✓ 채택
PAY_3	100.0%	✓ 채택
PAY_4	100.0%	✓ 채택
BILL_AMT1	100.0%	✓ 채택
PAY_AMT1	100.0%	✓ 채택
PAY_AMT2	100.0%	✓ 채택
PAY_AMT3	100.0%	✓ 채택
PAY_AMT6	100.0%	✓ 채택
PAY_5	98.8%	✓ 채택
BILL_AMT5	98.8%	✓ 채택
PAY_6	97.5%	✓ 채택
PAY_AMT4	96.2%	✓ 채택
PAY_AMT5	95.0%	✓ 채택
SEX	91.2%	✓ 채택
BILL_AMT6	88.8%	✓ 채택
BILL_AMT2	71.2%	✓ 채택
BILL_AMT3	56.2%	제외

Feature	Selection Frequency	채택
BILL_AMT4	28.7%	제외
EDUCATION	21.2%	제외
MARRIAGE	15.0%	제외